

**CST8508 – Machine Vision Midterm Test****Test Duration: 60 minutes****Total Marks: 25**

First Name: _____

Last Name: _____

Student Number: _____

Please read following instructions carefully before starting your exam.**INSTRUCTIONS**

- Fill in your name and student Number in the designated area above.
- This test contributes **15% to your final grade.**
- Provide answers in the designated space in this sheet
- No paper is permitted to leave the room, all paper used, including scrap paper, must be returned before you exit the room.
- Calculators are allowed, however other personal electronic devices are **not permitted** in the classroom during the time of test.

Multiple Choice Questions

(20 x .5 = 10 Marks)

1. What is the image processing technique that enhance the image by altering the pixels?
 - a. Rotation
 - b. Translation
 - c. **Filtering**
 - d. Resizing
2. Which filtering technique increases the contrast between adjacent pixels to highlight details?
 - a. Blurring
 - b. **Sharpening**
 - c. Smoothing
 - d. Erosion
3. What are morphological operations in image processing typically used for?
 - a. Color corrections
 - b. Image compression
 - c. Enhancing image resolution
 - d. **Processing images based on shapes**
4. What is a pixel in the context of an image?
 - a. A type of image file format
 - b. A vector graphics unit
 - c. **A numerical representation at a location in an image**
 - d. A unit used in image processing
5. What is the purpose of feature descriptors in image processing?
 - a. To enhance image quality
 - b. **To provide unique representation of detected features**
 - c. To compress images based on detected features
 - d. To detect features
6. What is the full form of SIFT in feature detection?
 - a. Scalable Image Filtering Technique
 - b. **Scale Invariant Feature Transform**
 - c. Scale Invariant Filter Transformation
7. Which technique is effective for human detection in computer vision?
 - a. **HOG**
 - b. SURF

- c. FAST
 - d. SIFT
8. What is the purpose of data augmentation in image classification?
- a. To reduce the size of the dataset
 - b. To expand the training dataset
 - c. To convert images to grayscale
 - d. To normalize pixel values
9. Which activation function is suitable for binary classification?
- a. Sigmoid
 - b. ReLU
 - c. Tanh
 - d. Softmax
10. What is the primary purpose of segmentation in image processing?
- a. To enhance image quality
 - b. To extract objects from an image
 - c. To detect edges
 - d. To compress images
11. In which industry is machine vision used for quality checks on assembly lines?
- a. Retail
 - b. Manufacturing
 - c. Entertainment
 - d. Healthcare
12. Which OpenCV function is used to identify contours in an image?
- a. `cv2.findContours()`
 - b. `cv2.detectContours()`
 - c. `cv2.locateContours()`
 - d. `cv2.searchContours()`
13. What was a significant turning point in the evolution of machine vision?
- a. The invention of digital cameras
 - b. The rise of deep learning techniques
 - c. The development of CCD sensors
 - d. The introduction of color spaces
14. What measures the directional changes in the intensity of an image?
- a. Image Histogram
 - b. Image Thresholding

- c. Image Gradient
 - d. Image orientation
15. Which activation function mitigates vanishing gradient problem?
- a. Sigmoid
 - b. ReLU
 - c. Tanh
 - d. Softmax
16. What is the output of segmentation?
- a. RGB image
 - b. Grayscale image
 - c. Binary image
 - d. Cropped image
17. What is the process of labeling images with class names?
- a. Data Loading
 - b. Data Augmentation
 - c. Data Annotation
 - d. Data transformation
18. What does the horizontal axis of an ROC curve represent?
- a. True Positive rate
 - b. False Positive rate
 - c. True Negative rate
 - d. False Negative rate
19. Which function quantifies the difference between model predictions and true outcome?
- a. Convolution
 - b. Optimizer
 - c. Loss Function
 - d. Activation Function
20. Which technique can be used for achieving segmentation?
- a. Contours
 - b. Thresholding
 - c. Filtering
 - d. Smoothing

Fill in the blanks Questions and Short answer Questions

(10 x 1 = 10 Marks)

21. The _____ thresholding applies two thresholds (high and low) to classify edges into strong, weak and non-edges in Canny edge detection?

Answer: **Double**

22. A _____ is a curve that joins a set of points enclosing an area having same colour or intensity?

Answer: **Contour**

23. Hidden Layers in CNN consists of convolutional layers, _____ layers and fully connected layers?

Answer: **Pooling**

24. _____ measures the proportion of total predictions (both positive and negative) that the model got correct?

Answer: **Accuracy**

25. _____ helps to preserve edge information during convolution operation?

Answer: **Padding**

26. What are two examples of learnable parameters in a CNN?

Answer: **weights and bias**

27. What is the process called when dilation is followed by erosion?

Answer: **closing**

28. What is stride in convolution operation?

Answer: **Step size with which filter slides across input image**

29. What is the full form of ORB?

Answer: **Oriented FAST and rotated BRIEF**

30. Neurons in the last layer match number of _____, in a classification model?

Answer: **Classes**

Mathematical Question

31. Given a grayscale image of size 32×32 pixels, you apply a convolutional layer with 16 filters, each of size 5×5 , with a stride of 1 and padding = 0. What will be the dimensions of the output feature map?
(2 Marks)

32. Using the given matrix A and kernel K, perform the convolution operation with stride 1 and no padding and compute the resultant matrix? (3 Marks)

Matrix A:

1	5	3	9	0
5	9	6	1	2
7	1	9	0	1
8	2	3	4	8
6	7	8	5	9

Kernel K:

-2	0	2
-2	0	2
-2	0	2

Answer:

